



UNIVERSITY OF
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Superconductivity

Lecture 03 – “London Brothers”

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Previously,

Key observation:

- $\rho = 0$ ($E = 0$),
- $B = 0$,

with some penetration depth λ .

Electromagnetism

- Maxwell's equations
- Ohm's law ($V = IR$ or $\mathbf{j} = \sigma \mathbf{E}$): valid when it's valid ...

We will address thermodynamics tomorrow.

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No scattering

Electron transport

An electron inside a metal may respond (move) to an electric field:

$$m_e \left(\frac{d\mathbf{v}}{dt} + \frac{\mathbf{v}}{\tau} \right) = e\mathbf{E}. \quad (1)$$

On the left,

- 1st term: acceleration
- 2nd term: drag due to various scattering (by defects, other electrons or phonons...)

This simple model is called the Drude model.

For a perfect conductor, $\tau \rightarrow \infty$, using the definition of the current density, $\mathbf{j} = nev$, one obtains,

$$\frac{d\mathbf{j}}{dt} = \frac{ne^2}{m_e} \mathbf{E}, \quad (2)$$

$\Rightarrow$ To maintain a constant current, $E = 0$

Electron transport - B field version (1/3)

Let us find out the equation's implication for a magnetic field:
Starting from a London equation,

$$\frac{d\mathbf{j}}{dt} = \frac{ne^2}{m_e} \mathbf{E}, \quad (3)$$

Taking the curl on both sides gives

$$\nabla \times \frac{d\mathbf{j}}{dt} = \frac{ne^2}{m_e} (\nabla \times \mathbf{E}). \quad (4)$$

Using the Faraday's law of induction, $\nabla \times \mathbf{E} = -d\mathbf{B}/dt$, the above becomes,

$$\frac{d}{dt} (\nabla \times \mathbf{j}) = -\frac{ne^2}{m_e} \frac{d\mathbf{B}}{dt} \quad (5)$$

Electron transport - B field version (2/3)

$$\frac{d}{dt} (\nabla \times \mathbf{j}) = -\frac{ne^2}{m_e} \frac{d\mathbf{B}}{dt} \quad (6)$$

Using the Ampere's law, $\nabla \times \mathbf{B} = \mu_0 \mathbf{j}$, one obtains,

$$\frac{1}{\mu_0} \frac{d}{dt} [\nabla \times (\nabla \times \mathbf{B})] = -\frac{ne^2}{m_e} \frac{d\mathbf{B}}{dt} \quad (7)$$

One may further use

- Vector identity: $\nabla \times (\nabla \times \mathbf{B}) = \nabla (\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B}$
- Maxwell's no monopole: $\nabla \cdot \mathbf{B} = 0$,

to obtain, finally,

$$\frac{d}{dt} \left[\nabla^2 \mathbf{B} - \frac{\mu_0 ne^2}{m_e} \mathbf{B} \right] = 0. \quad (8)$$

Electron transport - B field version (3/3)

What's the meaning of the equation,

$$\frac{d}{dt} \left[\nabla^2 \mathbf{B} - \frac{\mu_0 n e^2}{m_e} \mathbf{B} \right] = 0. \quad (9)$$

One is motivated to define a length scale λ_L as

$$\frac{\mu_0 n e^2}{m_e} = \frac{1}{\lambda_L^2}. \quad (10)$$

Then the equation can be rewritten, for the simplest case with a single coordinate x , as

$$\frac{\partial^2}{\partial x^2} \left(\frac{\partial \mathbf{B}}{\partial t} \right) = \frac{1}{\lambda_L^2} \left(\frac{\partial \mathbf{B}}{\partial t} \right). \quad (11)$$

and its solution,

$$\frac{\partial \mathbf{B}}{\partial t} = \frac{\partial \mathbf{B}}{\partial t} \Big|_{x=0} \exp(-x/\lambda_L). \quad (12)$$

⇒ Regardless how one may change the applied field from outside a perfect conductor, nothing really would happen inside.

Today, We have so far concerned with a perfect conductor, not necessarily a superconductor.

Meissner & Ochsenfeld

Penetration depth

How to make

$$\left[\nabla^2 \mathbf{B} - \frac{\mu_0 n e^2}{m_e} \mathbf{B} \right] = \text{const.} \quad (13)$$

compatible with the experimental observation of field penetration?

Simply,

$$\left[\nabla^2 \mathbf{B} - \frac{\mu_0 n e^2}{m_e} \mathbf{B} \right] = 0, \quad (14)$$

to obtain

$$B = B_a \exp(-x/\lambda_L), \quad (15)$$

where

$$\frac{\mu_0 n e^2}{m_e} = \frac{1}{\lambda_L^2}. \quad (16)$$

⇒ More super-electrons (larger n), less magnetic field penetration.

Penetration depth

You may trace back where the last assumption might have entered in our derivation, which is

$$\frac{d}{dt} (\nabla \times \mathbf{j}) = -\frac{ne^2}{m_e} \frac{d\mathbf{B}}{dt}, \quad (17)$$

where you remove the time derivative from both sides, then obtain

$$\nabla \times \mathbf{j} = -\frac{ne^2}{m_e} \mathbf{B}, \quad (18)$$

which is the other London equation.

London equations

Using λ_L , neatly written as

$$\nabla \times \mathbf{j} = -\frac{1}{\mu_0 \lambda_L^2} \mathbf{B}, \quad (19)$$

$$\frac{\partial \mathbf{j}}{\partial t} = \frac{1}{\mu_0 \lambda_L^2} \mathbf{E}, \quad (20)$$

corresponding to perfect expulsion (Meissner effect) and zero resistance (perfect conductivity), respectively.

An alternative form

London equation is often expressed as

$$\mathbf{j}(\mathbf{r}) = -\frac{ne^2}{m_e}\mathbf{A}(\mathbf{r}) \quad (21)$$

Assumptions

So far we have implicitly assumed that

- Response of an electron to a field is local,
- DC currents only.

Non-local response

Generalising the London equations,

$$\mathbf{j}(\mathbf{r}) = -\frac{ne^2}{m_e} \frac{3}{4\pi\xi_0} \int \frac{\mathbf{R}(\mathbf{R} \cdot \mathbf{A}(\mathbf{r}'))}{R^4} e^{-R/\xi} d\mathbf{r}' \quad (22)$$

where $\mathbf{R} = \mathbf{r} - \mathbf{r}'$ and

$$\frac{1}{\xi} = \frac{1}{\xi_0} + \frac{1}{l}, \quad (23)$$

where ξ is called the coherence length and l is the mean free path of the electrons.

In the local limit, it reduces to

$$\mathbf{j}(\mathbf{r}) = -\frac{\xi}{\xi_0} \frac{ne^2}{m_e} \mathbf{A}(\mathbf{r}). \quad (24)$$

Coherence length estimate

From the uncertainty principle, $\Delta x \Delta p \gtrsim \hbar$, and the argument that only those electrons within $k_B T_c$ of the Fermi energy playing a major role, one obtains,

$$\xi_0 \sim \frac{\hbar v_F}{k_B T_c}. \quad (25)$$